

Subject: Elective (Embedded System Design)

Class: B. Tech

Semester: VIII

Date: _____

Time: 3 hr.

Max. Marks: 70

INSTRUCTIONS: 1. Q.1 is Compulsory.

2. Attempt any Five question from Q.2 to Q.7.

3. Sub question from each main question must be solved together

Q. 1 Fill in the blanks

(10M)

- i. CAN bus protocol, using standard frame format identifier isbits.
- ii. LPC 2148 have of on chip flash.
- iii. ARM 7 support stage pipeline.
- iv. register is used to access data memory indirectly in PIC 16F84A.
- v. A is a key that your code acquires in order to continue execution.

Q.2 A) Explain following design metrics and its significance.

(06M)

- i) NRE cost;
- ii) Time to market;
- iii) Power

B) Compare RISC and CISC instruction set architecture.

(06M)

Q.3 A) Give the feature of DSP processor. how they are differ from normal processor?

(06M)

B) Draw REGISTER FILE MAP OF PIC 16F84A.

(06M)

Q.4 A) Write assembly language program for PIC 16F84A to add 5 no present in Data memory

starting at 0X25 H and store the result in 0X40 H.

(06M)

B) Which design metrics will be given more priority for designing "Weather monitoring system in remote area".

(06M)

(P.T.O.)

Q.5 A) What are the feature of CAN protocol? explain bus arbitration. (06M)

B) Can optimization code be used in embedded C++ programs ? justify. (06M)

Q.6 A) Give suitable example where RMA scheduling fails but DMA will succeed.

explain.(06M)

B) what are the situations, which lead to priority inversion problems? How does an OS solve this problem by a priority inheritance mechanism? (06M)

Q.7 A) what are the different method to have exclusive access to shared resource? Explain.(06M)

B) Why you need a cross compiler? why use an infinite loop in embedded software? (06M)

*****BEST OF LUCK*****